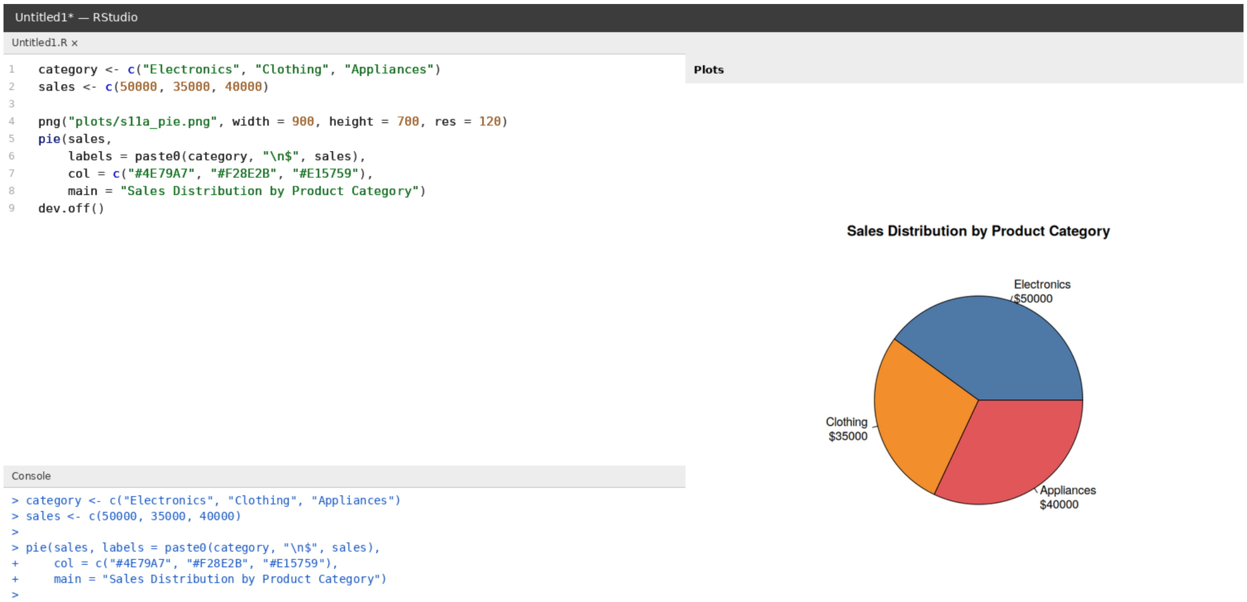
List of Experiments — Implementation Screenshots

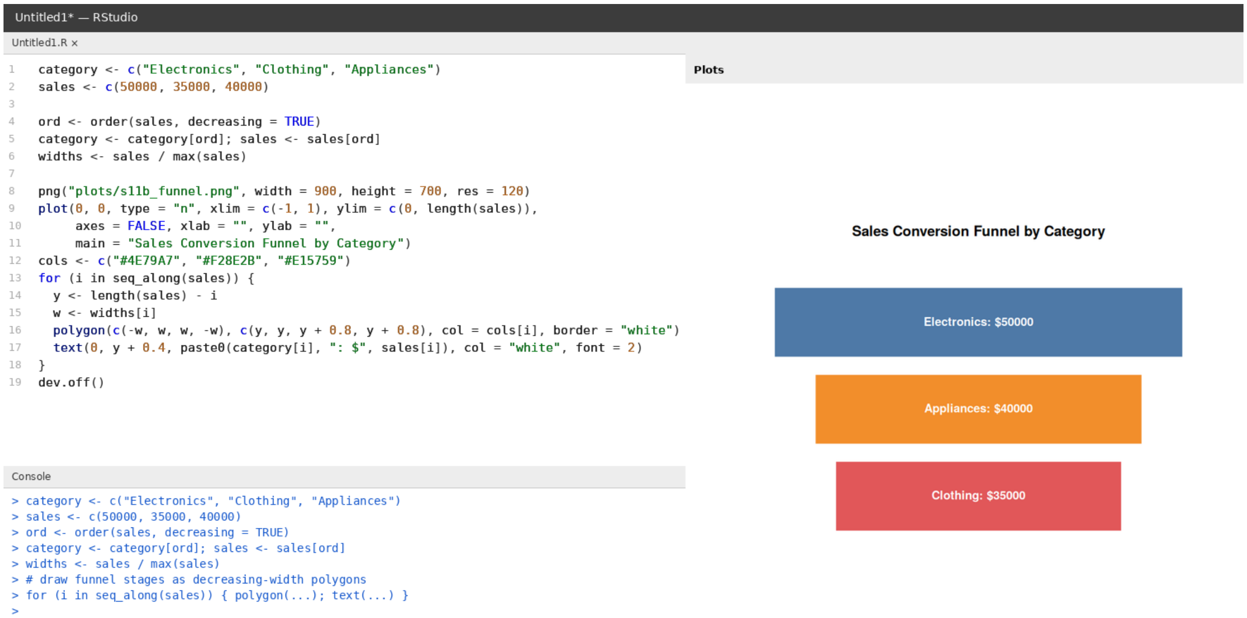
Roshan R (192524085)

Set 11 — Product Category Analysis

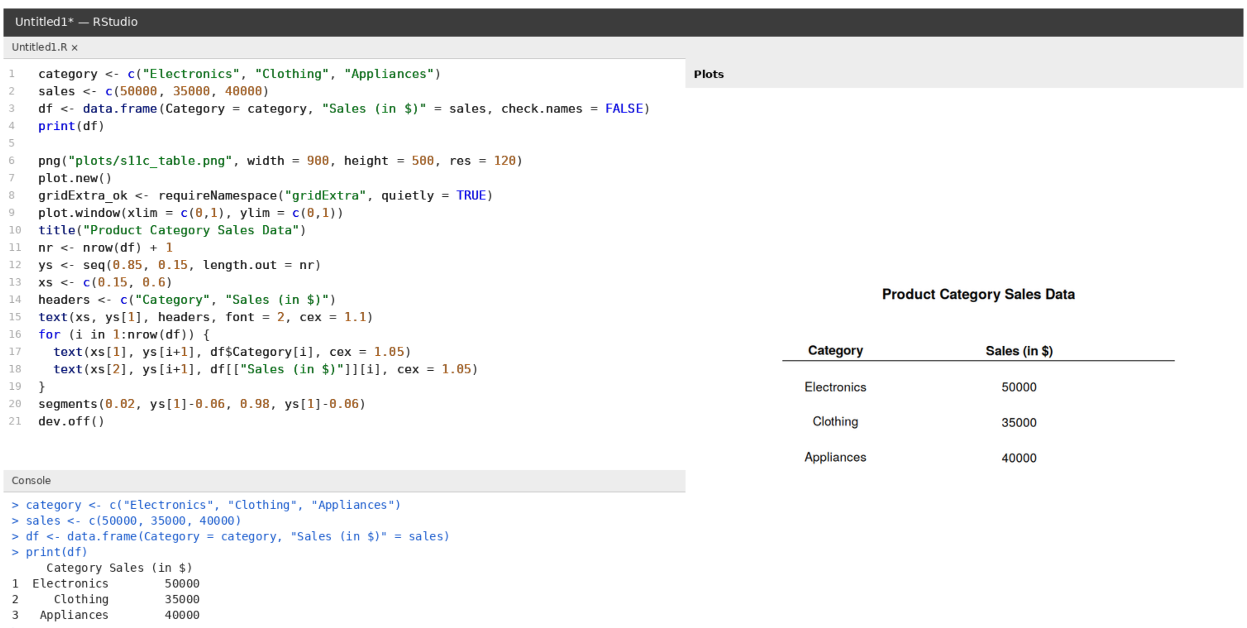
1. Create a pie chart to represent the distribution of sales across product categories. Include labels.



2. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.



3. Build a table to display the sales data for each product category. Label the table elements.



4. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

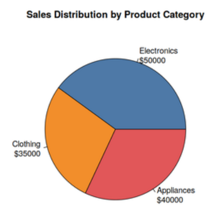
Tableau Public — Product Category Dashboard

File Data Format Dashboard Story Window Help

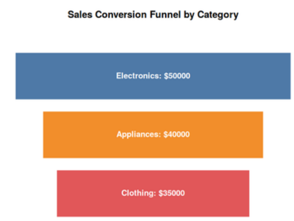
Sheets

- Sales by Category (Pie
- Category Conversion (F
- Category Sales Data (T

Sales by Category (Pie)



Category Conversion (Funnel)



Category Sales Data (Table)

Product Category Sales Data

Category	Sales (in $)
Electronics	50000
Clothing	35000
Appliances	40000

Set 12 — Website Traffic Analysis

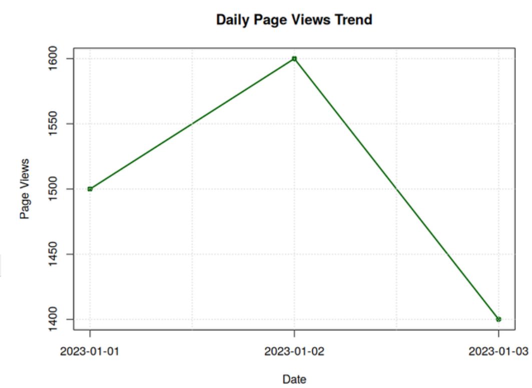
1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

Untitled1* — RStudio

Untitled1.R ×

```r
date <- c("2023-01-01", "2023-01-02", "2023-01-03")
page_views <- c(1500, 1600, 1400)

png("plots/s12a_line.png", width = 900, height = 700, res = 120)
plot(1:3, page_views, type = "o", col = "darkgreen", lwd = 2, pch = 16,
     xaxt = "n", xlab = "Date", ylab = "Page Views",
     main = "Daily Page Views Trend")
axis(1, at = 1:3, labels = date)
grid()
dev.off()
```

Plots



Console

```
> date <- c("2023-01-01", "2023-01-02", "2023-01-03")
> page_views <- c(1500, 1600, 1400)
>
> plot(1:3, page_views, type = "o", col = "darkgreen", lwd = 2,
+      pch = 16, xaxt = "n", xlab = "Date", ylab = "Page Views",
+      main = "Daily Page Views Trend")
> axis(1, at = 1:3, labels = date)
> grid()
>
```

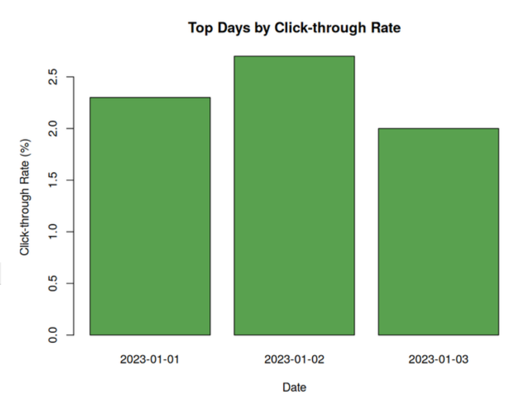
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

Untitled1* — RStudio

Untitled1.R ×

```r
date <- c("2023-01-01", "2023-01-02", "2023-01-03")
ctr <- c(2.3, 2.7, 2.0)

png("plots/s12b_bar.png", width = 900, height = 700, res = 120)
barplot(ctr, names.arg = date, col = "#59A14F",
        xlab = "Date", ylab = "Click-through Rate (%)",
        main = "Top Days by Click-through Rate")
dev.off()
```

Plots

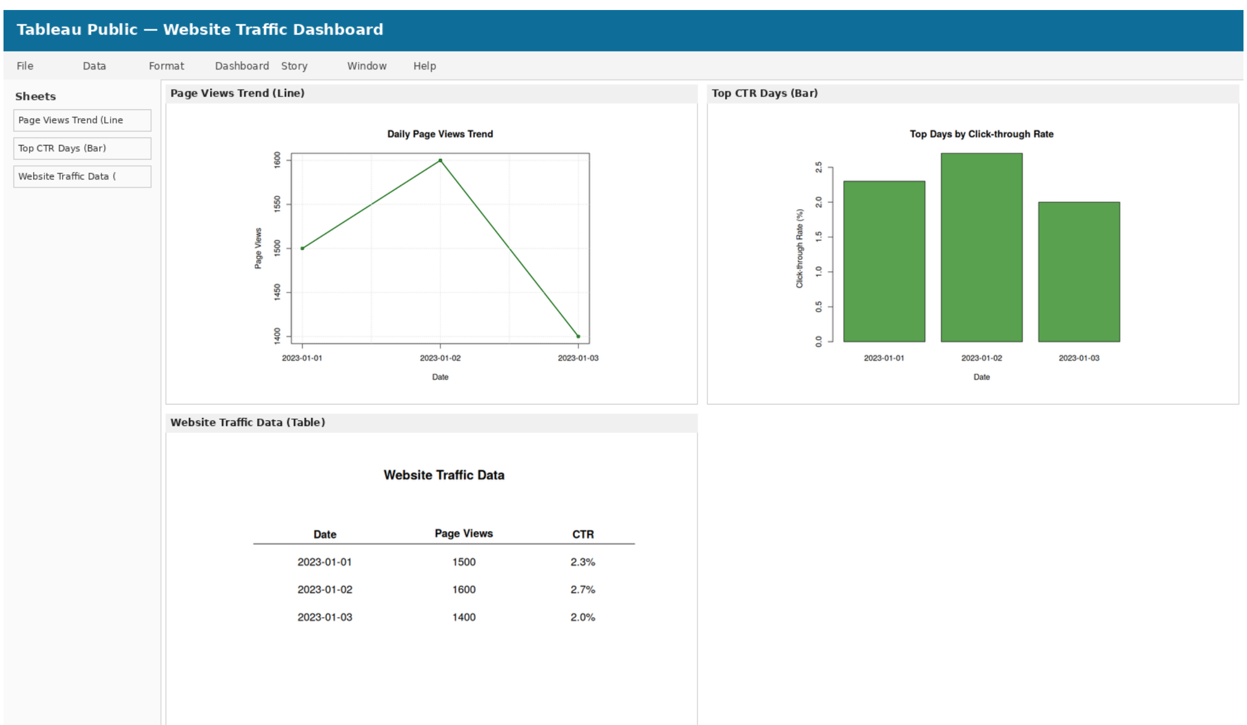


Console

```
> date <- c("2023-01-01", "2023-01-02", "2023-01-03")
> ctr <- c(2.3, 2.7, 2.0)
>
> barplot(ctr, names.arg = date, col = "#59A14F",
+         xlab = "Date", ylab = "Click-through Rate (%)",
+         main = "Top Days by Click-through Rate")
>
```

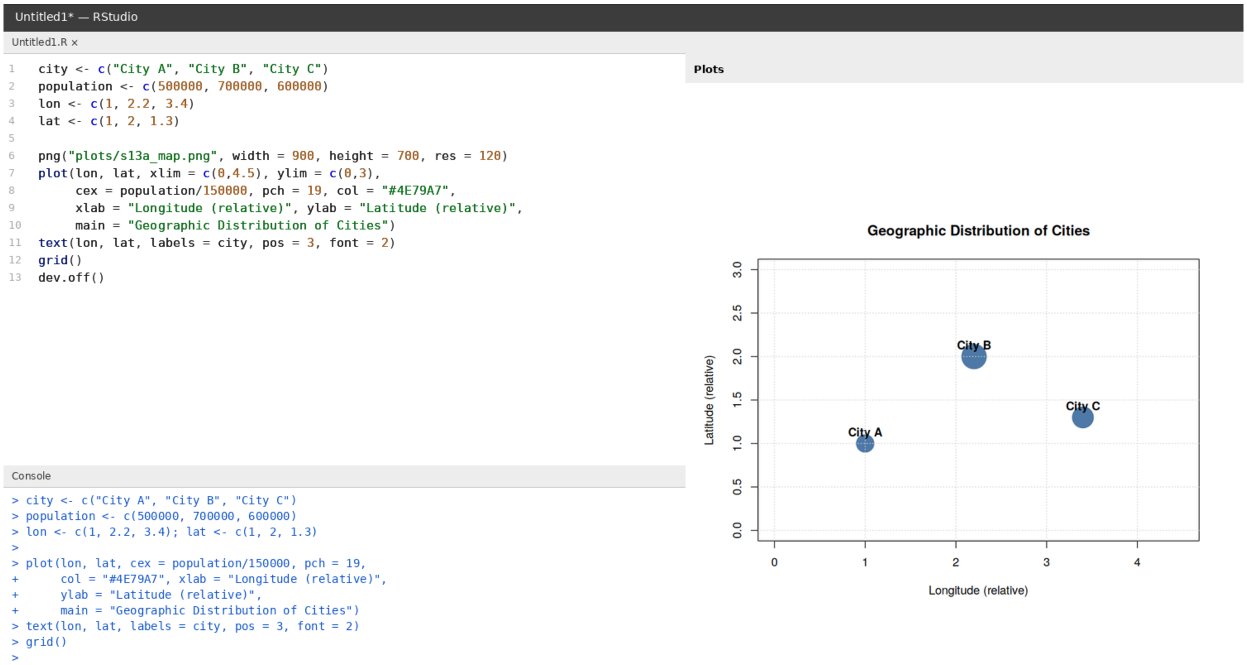
3. Build a table to show the website traffic data for each day. Label the table elements.

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

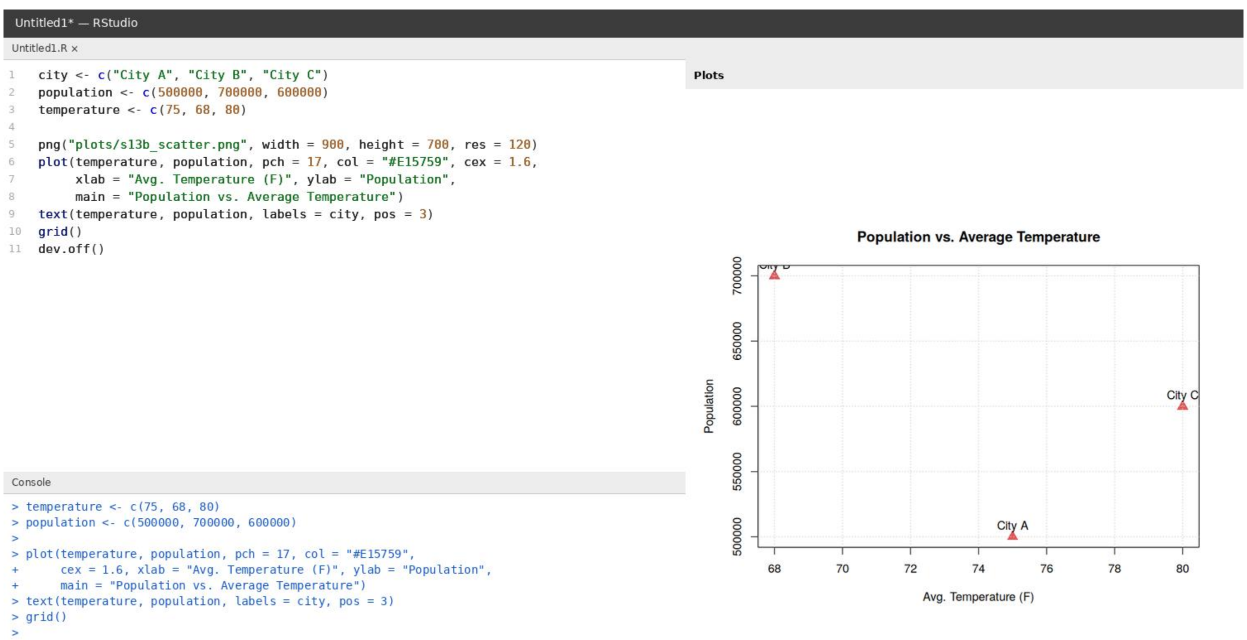


Set 13 — Geographic Data Analysis

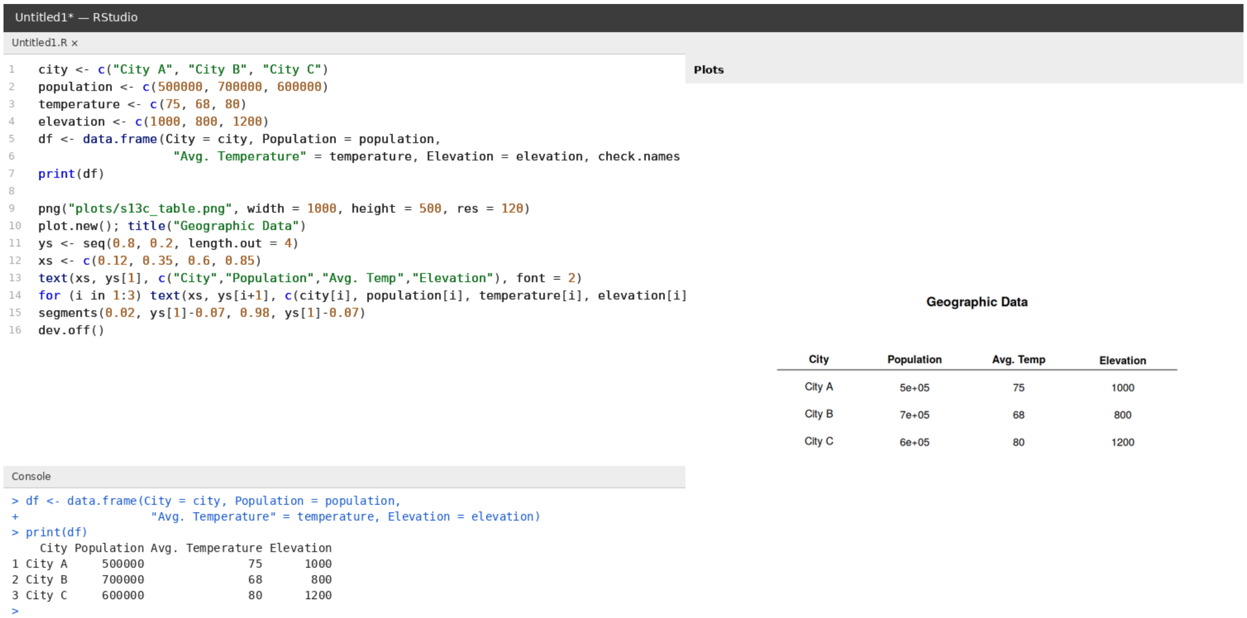
1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.



2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.



3. Build a table to display the geographic data for each city. Label the table elements.



4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Set 14 — Survey Results Analysis

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

Untitled1* — RStudio

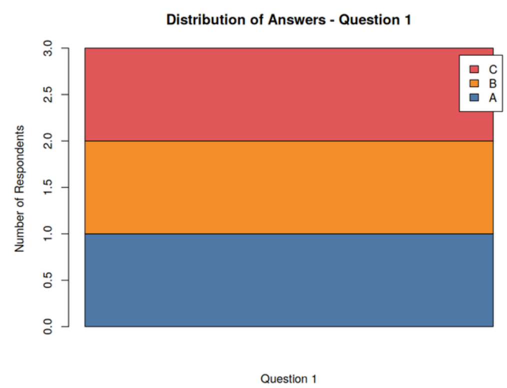
Untitled1.R ×

```r
resp <- c(1,2,3)
q1 <- c("A","B","C")

tab <- table(q1)
png("plots/s14a_stackedbar.png", width = 900, height = 700, res = 120)
barplot(as.matrix(tab), beside = FALSE, col = c("#4E79A7","#F28E2B","#E15759"),
        legend.text = rownames(tab), main = "Distribution of Answers - Question 1",
        ylab = "Number of Respondents", xlab = "Question 1")
dev.off()
```

Console

```
> resp <- c(1,2,3); q1 <- c("A","B","C")
> tab <- table(q1)
> barplot(as.matrix(tab), col = c("#4E79A7","#F28E2B","#E15759"),
+        legend.text = rownames(tab),
+        main = "Distribution of Answers - Question 1")
>
```

Plots



2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

Untitled1* — RStudio

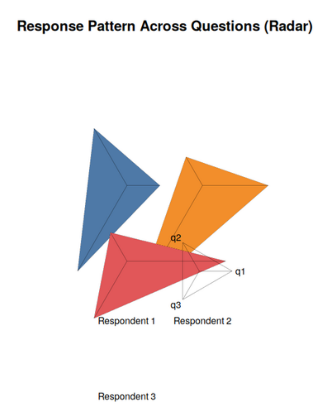
Untitled1.R ×

```r
library(graphics)
resp_names <- c("Respondent 1","Respondent 2","Respondent 3")
# map A/B/C/D to numeric score 1-4 for radar style comparison
map <- c(A=1,B=2,C=3,D=4)
q1 <- map[c("A","B","C")]
q2 <- map[c("B","A","A")]
q3 <- map[c("C","D","B")]
m <- rbind(q1, q2, q3)
colnames(m) <- resp_names

png("plots/s14b_radar.png", width = 900, height = 700, res = 120)
stars(t(m), scale = FALSE, full = TRUE, key.loc = c(4.5, 2),
      main = "Response Pattern Across Questions (Radar)",
      labels = resp_names, col.stars = c("#4E79A7","#F28E2B","#E15759"))
dev.off()
```

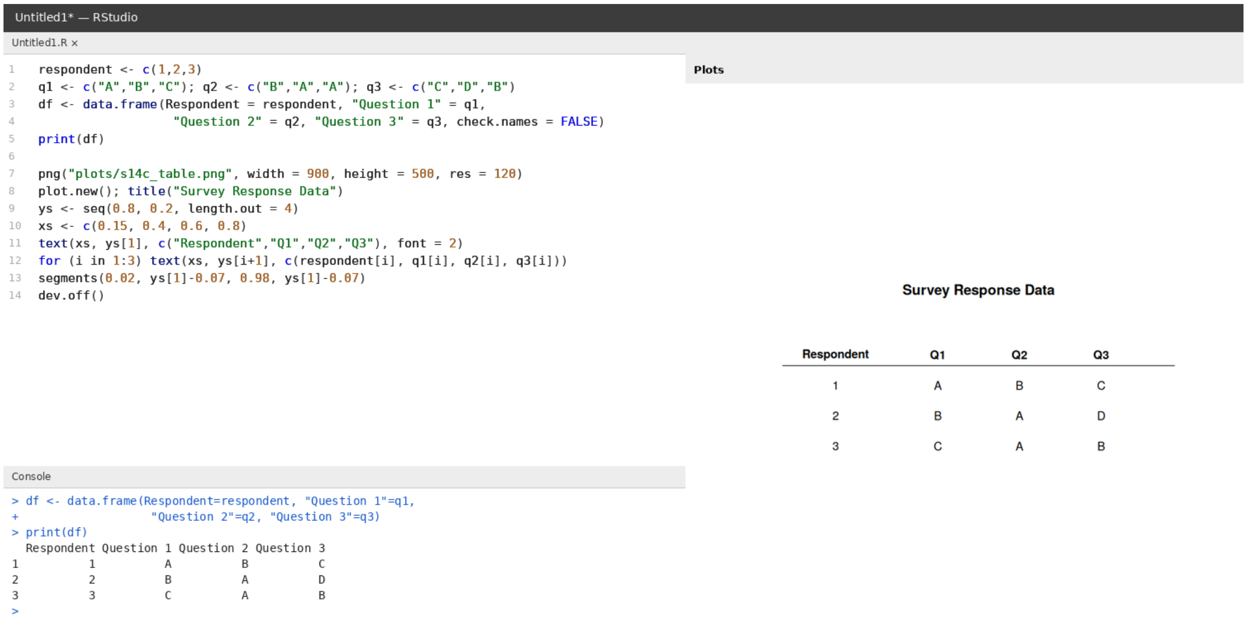
Console

```
> map <- c(A=1,B=2,C=3,D=4)
> q1 <- map[c("A","B","C")]; q2 <- map[c("B","A","A")]; q3 <- map[c("C","D","B")]
> m <- rbind(q1, q2, q3)
> stars(t(m), scale = FALSE, full = TRUE,
+      main = "Response Pattern Across Questions (Radar)")
>
```

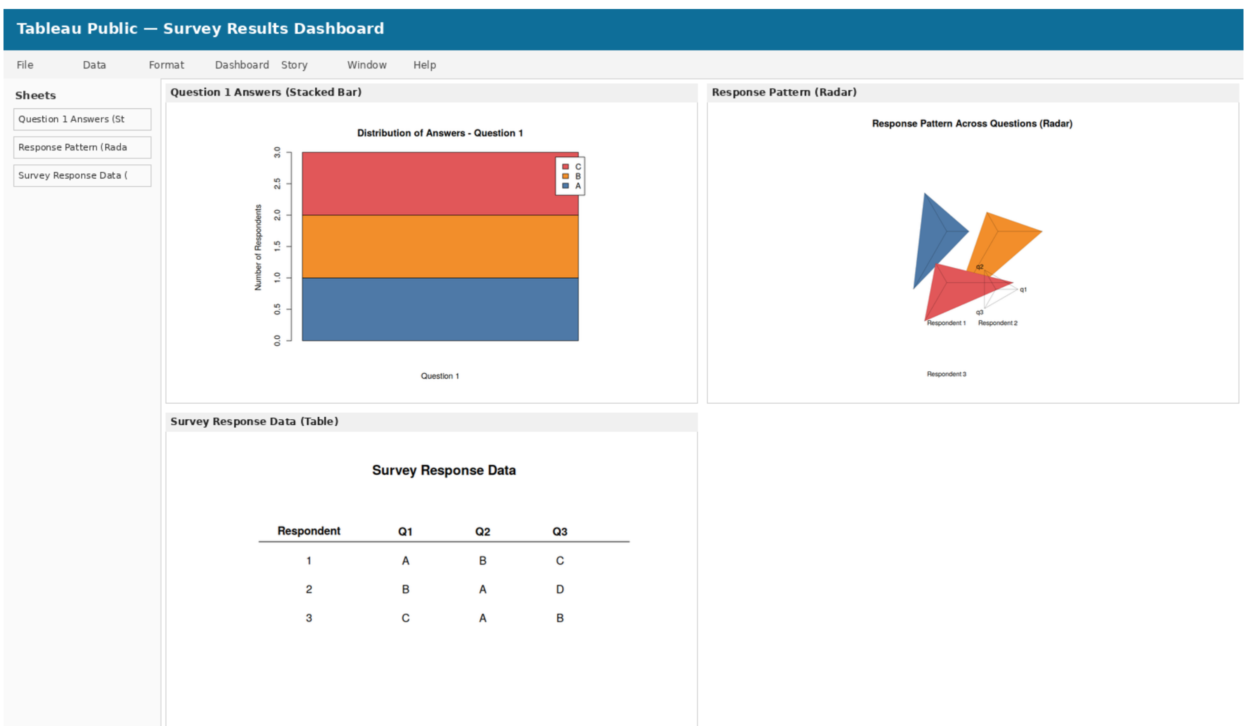
Plots



3. Build a table to show the survey response data for each respondent. Label the table elements.

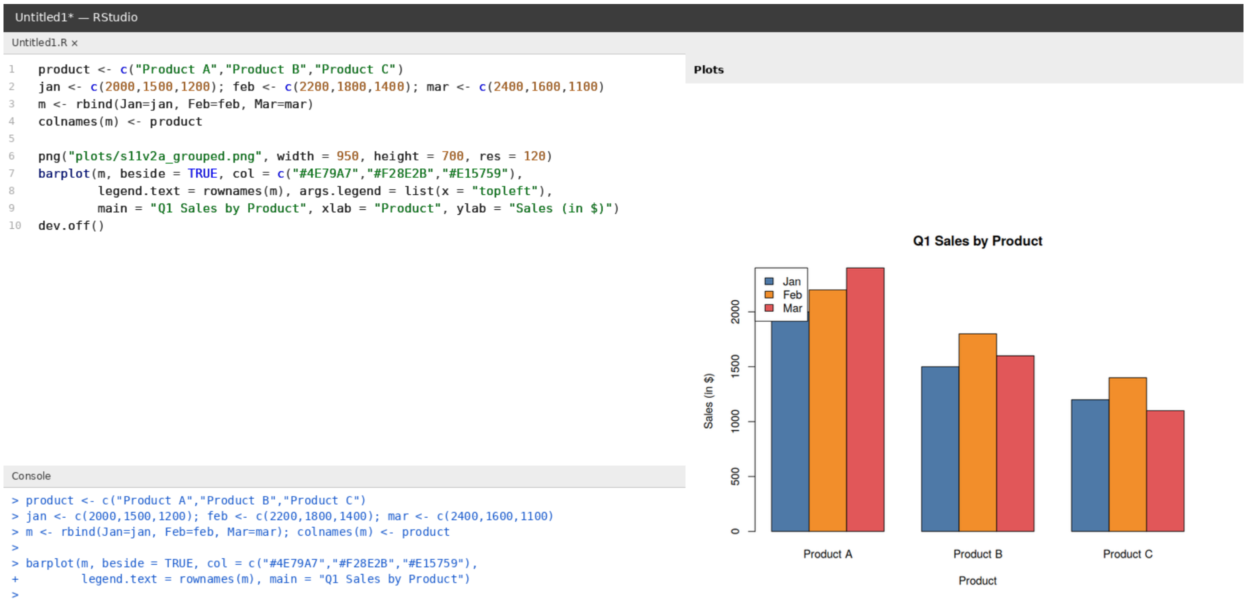


4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

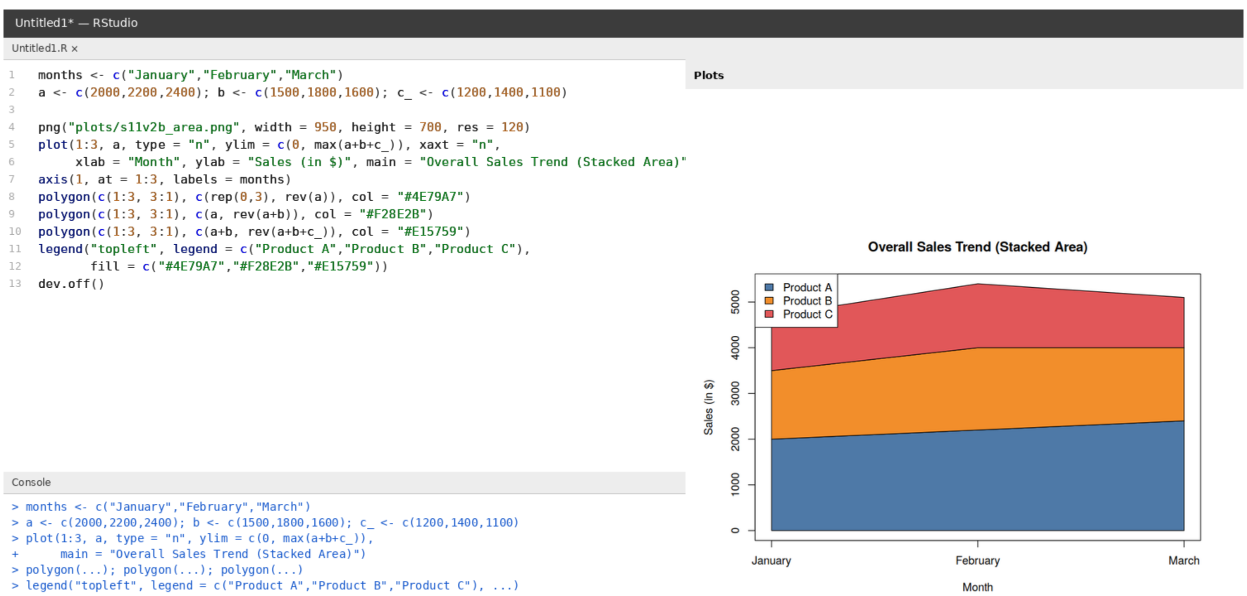


Set 11 — Product Sales Analysis

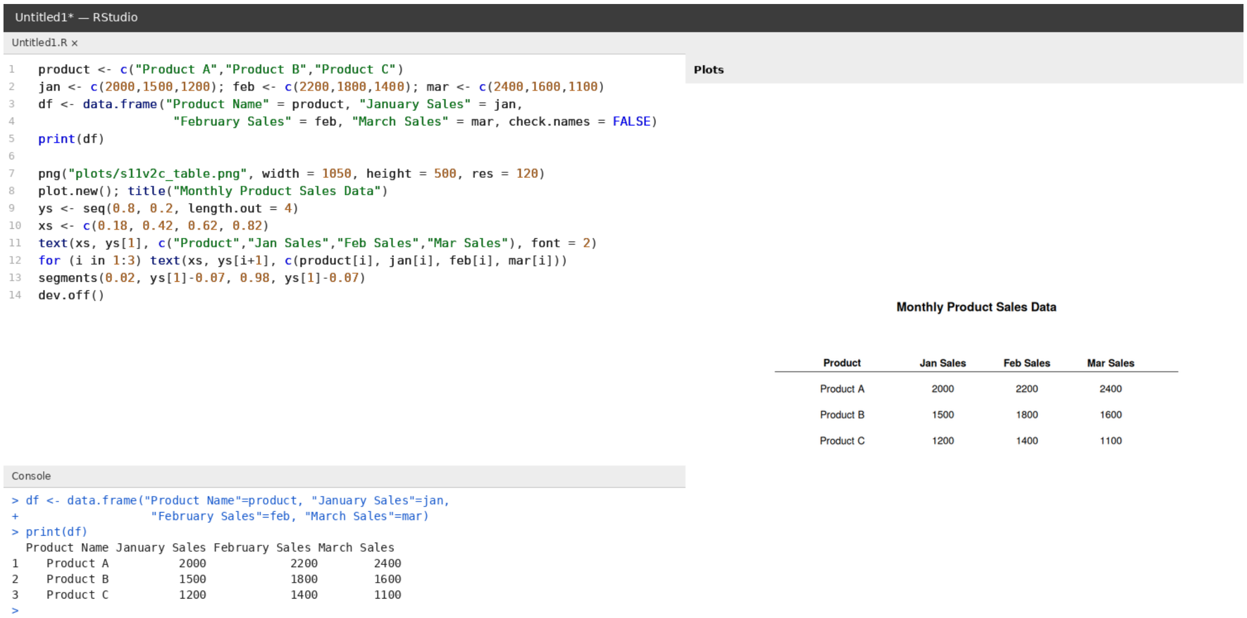
1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.



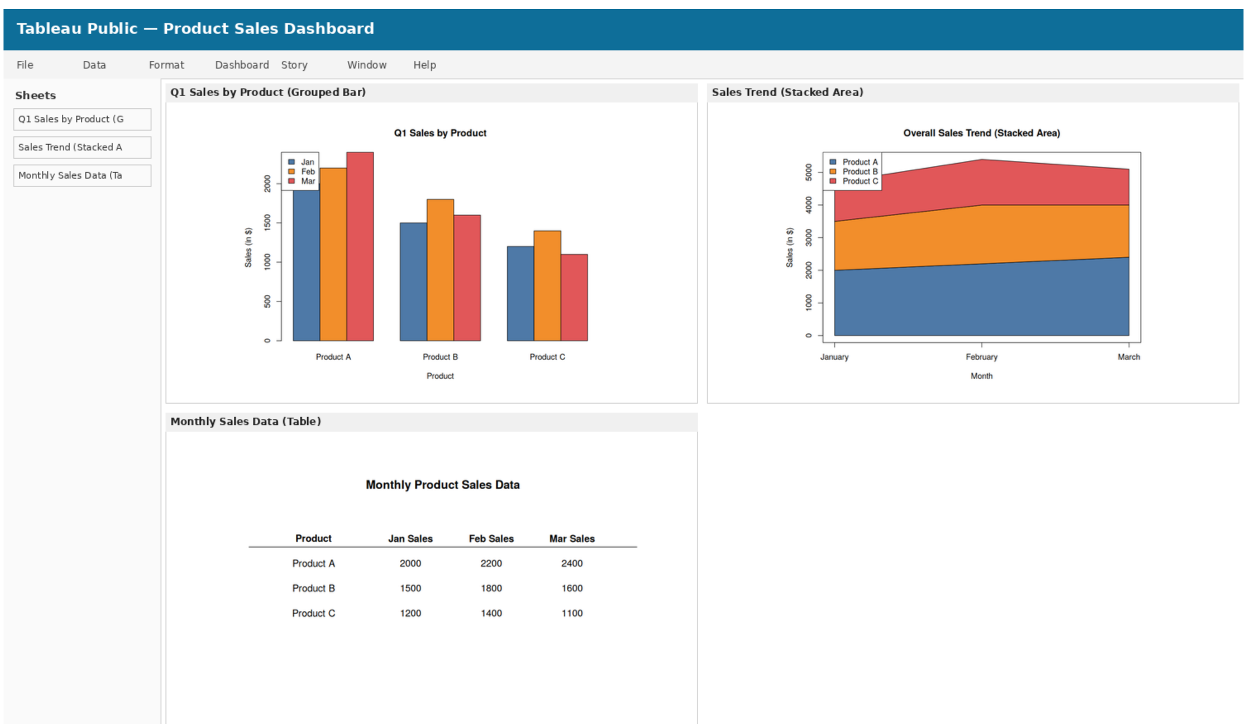
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.



3. Build a table to show the monthly sales data for each product. Label the table elements.

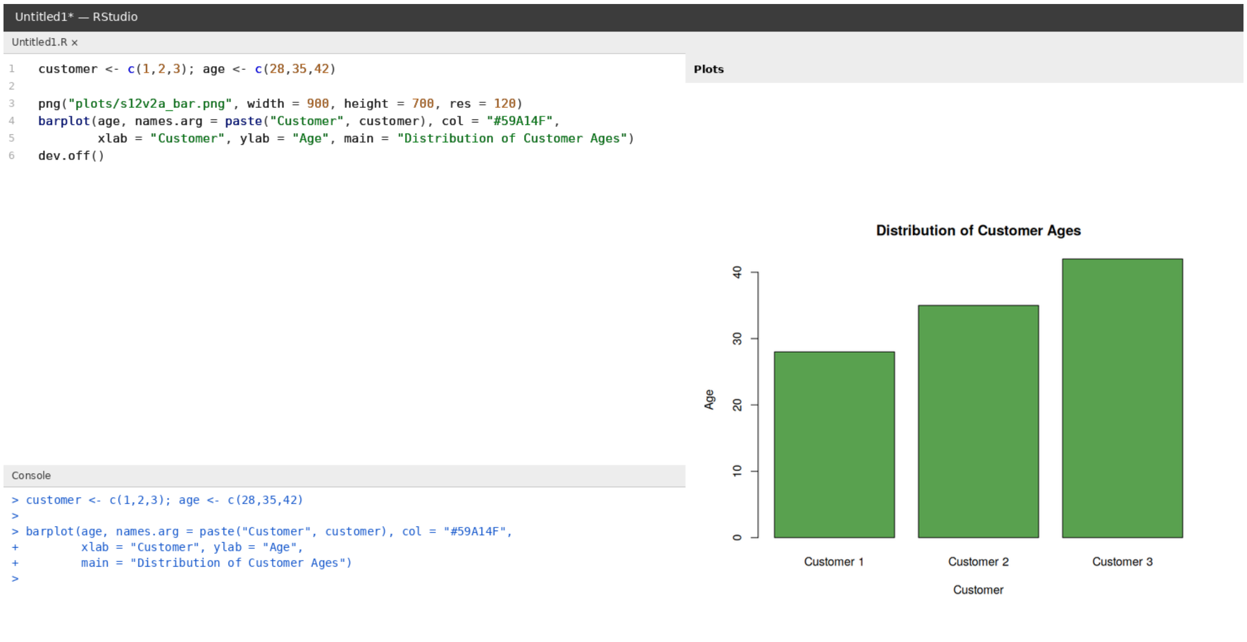


4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

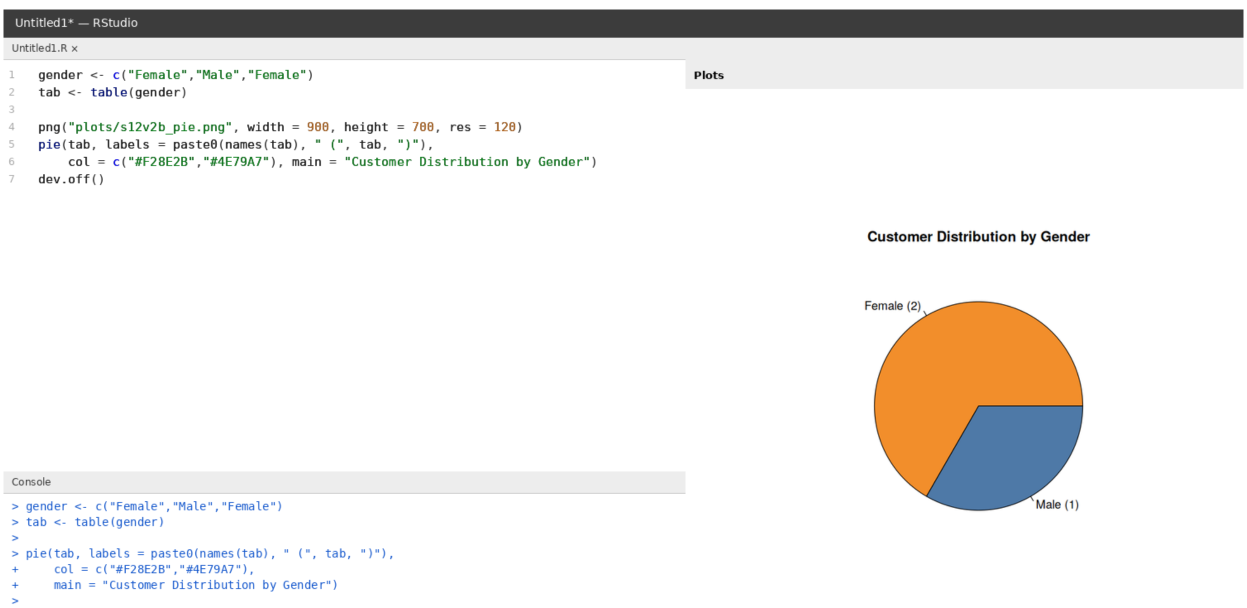


Set 12 — Customer Demographics Analysis

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

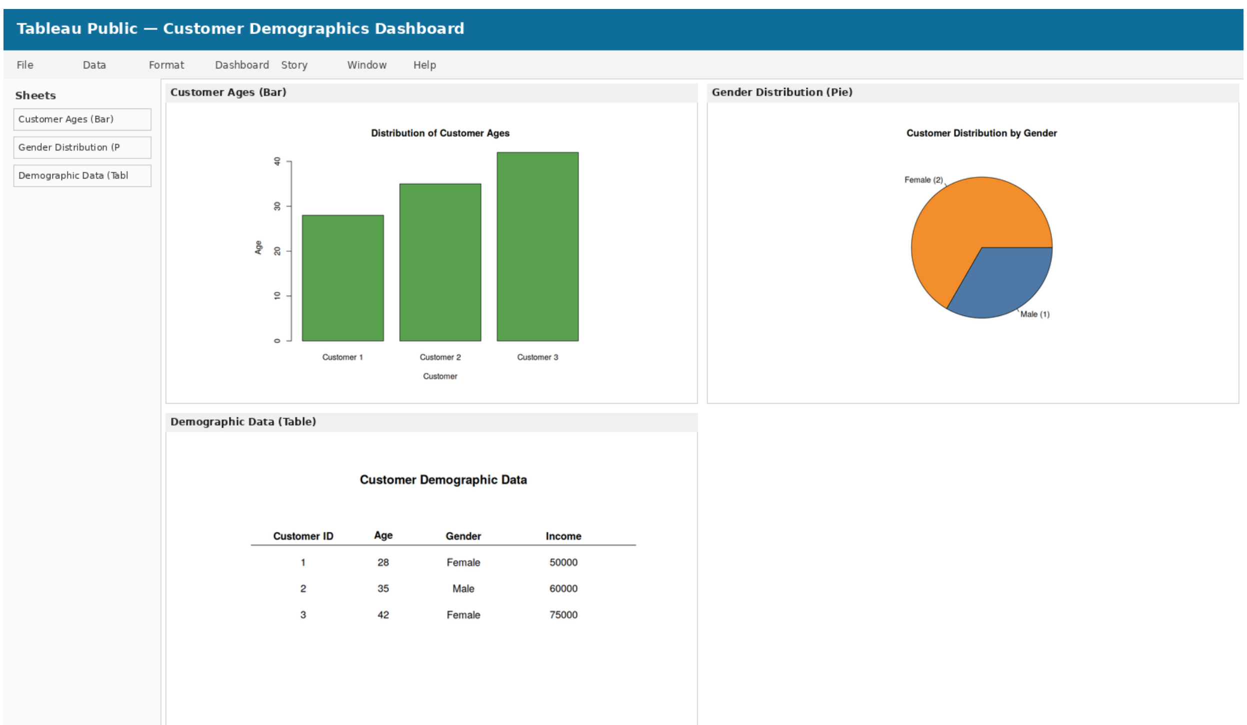


2. Generate a pie chart to represent the distribution of customers by gender.



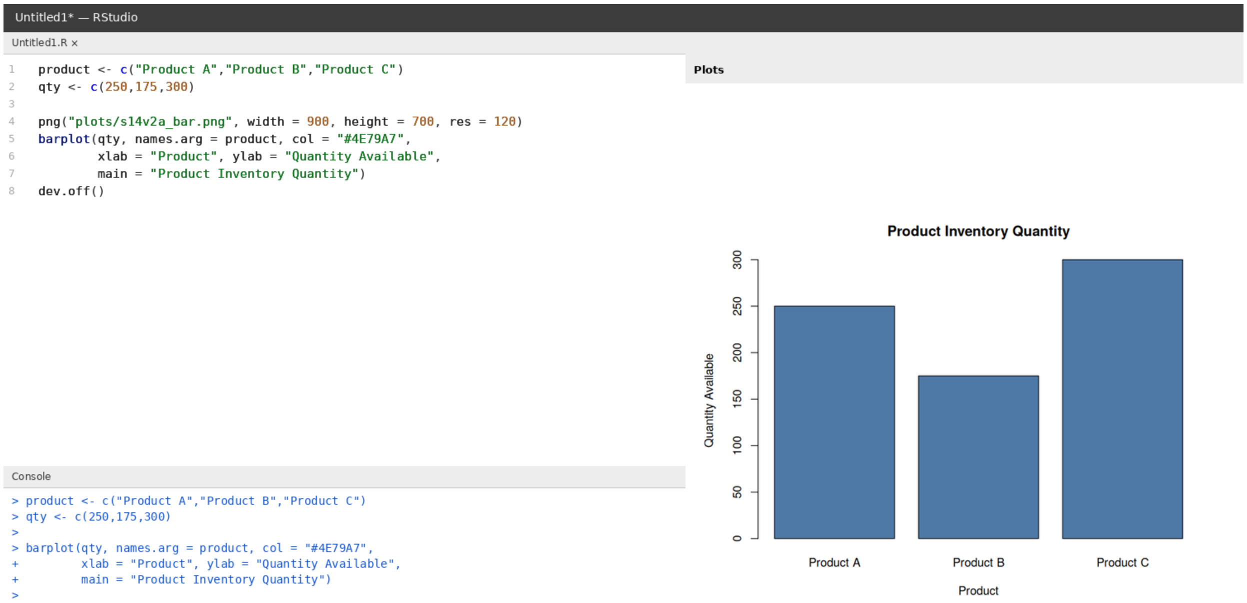
3. Build a table to show the demographic information for each customer. Label the table elements.

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

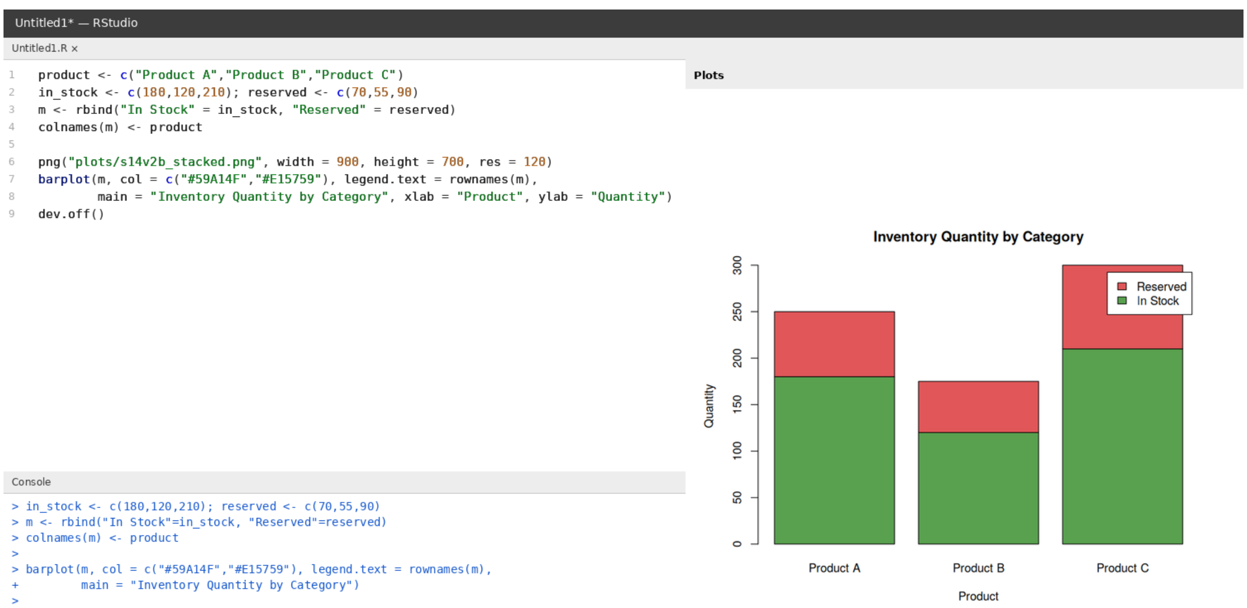


Set 14 — Product Inventory Management

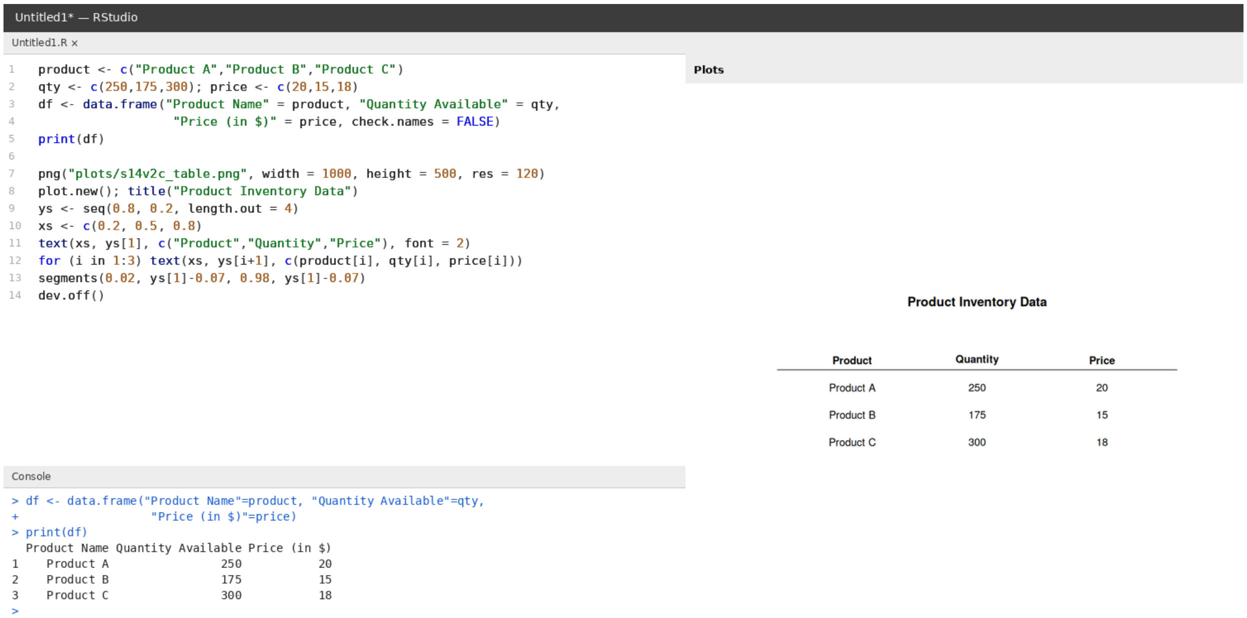
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.



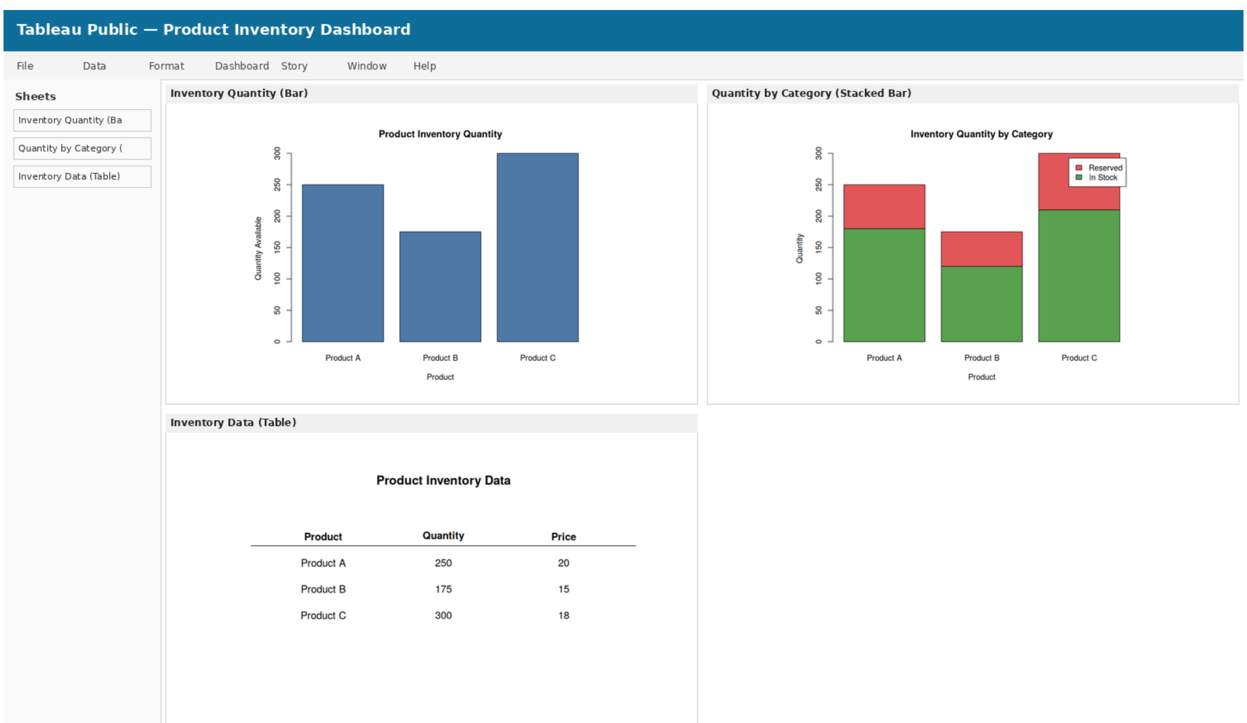
2. Generate a stacked bar chart to show the quantity of each product within different product categories.



3. Build a table to show the inventory data for each product. Label the table elements.



4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.



Set 15 — Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

Untitled1* — RStudio

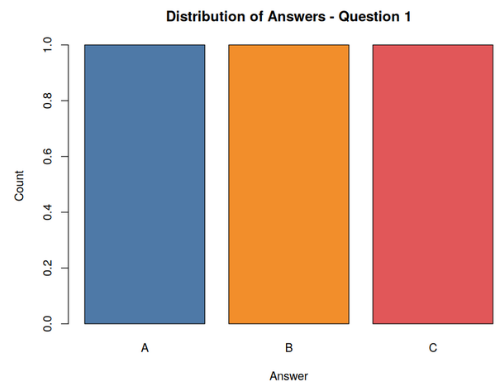
Untitled1.R ×

```r
survey_id <- c(1,2,3); q1 <- c("A","B","C")
tab <- table(q1)

png("plots/s15a_grouped.png", width = 900, height = 700, res = 120)
barplot(tab, col = c("#4E79A7","#F28E2B","#E15759"),
        xlab = "Answer", ylab = "Count",
        main = "Distribution of Answers - Question 1")
dev.off()
```

Console

```
> survey_id <- c(1,2,3); q1 <- c("A","B","C")
> tab <- table(q1)
>
> barplot(tab, col = c("#4E79A7","#F28E2B","#E15759"),
+         xlab = "Answer", ylab = "Count",
+         main = "Distribution of Answers - Question 1")
>
```

Plots



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

Untitled1* — RStudio

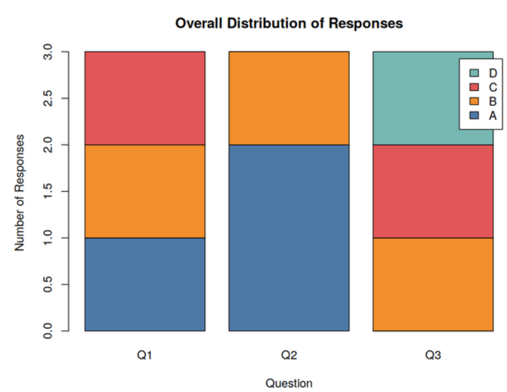
Untitled1.R ×

```r
q1 <- c("A","B","C"); q2 <- c("B","A","A"); q3 <- c("C","D","B")
opts <- c("A","B","C","D")
m <- rbind(Q1 = table(factor(q1, levels = opts)),
           Q2 = table(factor(q2, levels = opts)),
           Q3 = table(factor(q3, levels = opts)))

png("plots/s15b_stacked.png", width = 900, height = 700, res = 120)
barplot(t(m), col = c("#4E79A7","#F28E2B","#E15759","#76B7B2"),
        legend.text = opts, main = "Overall Distribution of Responses",
        xlab = "Question", ylab = "Number of Responses")
dev.off()
```

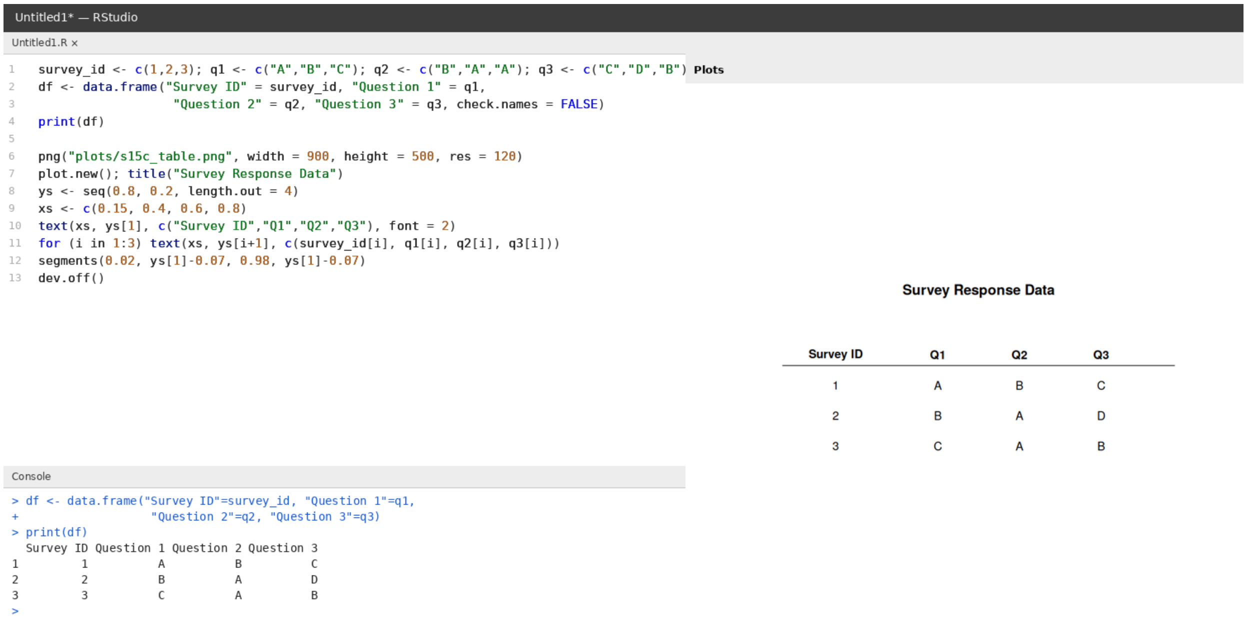
Console

```
> opts <- c("A","B","C","D")
> m <- rbind(Q1 = table(factor(q1, levels = opts)),
+            Q2 = table(factor(q2, levels = opts)),
+            Q3 = table(factor(q3, levels = opts)))
>
> barplot(t(m), col = c("#4E79A7","#F28E2B","#E15759","#76B7B2"),
+         legend.text = opts, main = "Overall Distribution of Responses")
>
```

Plots



3. Build a table to show the survey response data for each survey. Label the table elements.



4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Set 16 — Stock Analysis

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

Untitled1* — RStudio

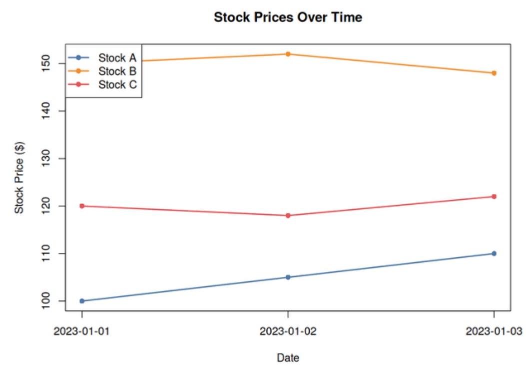
Untitled1.R ×

```r
date <- c("2023-01-01","2023-01-02","2023-01-03")
stockA <- c(100,105,110); stockB <- c(150,152,148); stockC <- c(120,118,122)

png("plots/s16a_line.png", width = 950, height = 700, res = 120)
plot(1:3, stockA, type = "o", col = "#4E79A7", lwd = 2, pch = 16,
     ylim = range(c(stockA,stockB,stockC)), xaxt = "n",
     xlab = "Date", ylab = "Stock Price ($)", main = "Stock Prices Over Time")
lines(1:3, stockB, type = "o", col = "#F28E2B", lwd = 2, pch = 16)
lines(1:3, stockC, type = "o", col = "#E15759", lwd = 2, pch = 16)
axis(1, at = 1:3, labels = date)
legend("topleft", legend = c("Stock A","Stock B","Stock C"),
       col = c("#4E79A7","#F28E2B","#E15759"), lwd = 2, pch = 16)
dev.off()
```

Console

```
> stockA <- c(100,105,110); stockB <- c(150,152,148); stockC <- c(120,118,122)
>
> plot(1:3, stockA, type = "o", col = "#4E79A7", ...)
> lines(1:3, stockB, type = "o", col = "#F28E2B", ...)
> lines(1:3, stockC, type = "o", col = "#E15759", ...)
> legend("topleft", legend = c("Stock A","Stock B","Stock C"), ...)
>
```

Plots



2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

Untitled1* — RStudio

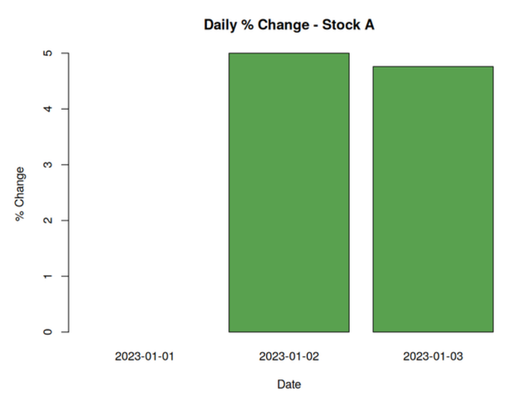
Untitled1.R ×

```r
stockA <- c(100,105,110)
pct_change <- round(c(NA, diff(stockA)/head(stockA,-1)*100), 2)
date <- c("2023-01-01","2023-01-02","2023-01-03")

png("plots/s16b_bar.png", width = 900, height = 700, res = 120)
barplot(pct_change, names.arg = date, col = "#59A14F",
        xlab = "Date", ylab = "% Change", main = "Daily % Change - Stock A")
dev.off()
```

Console

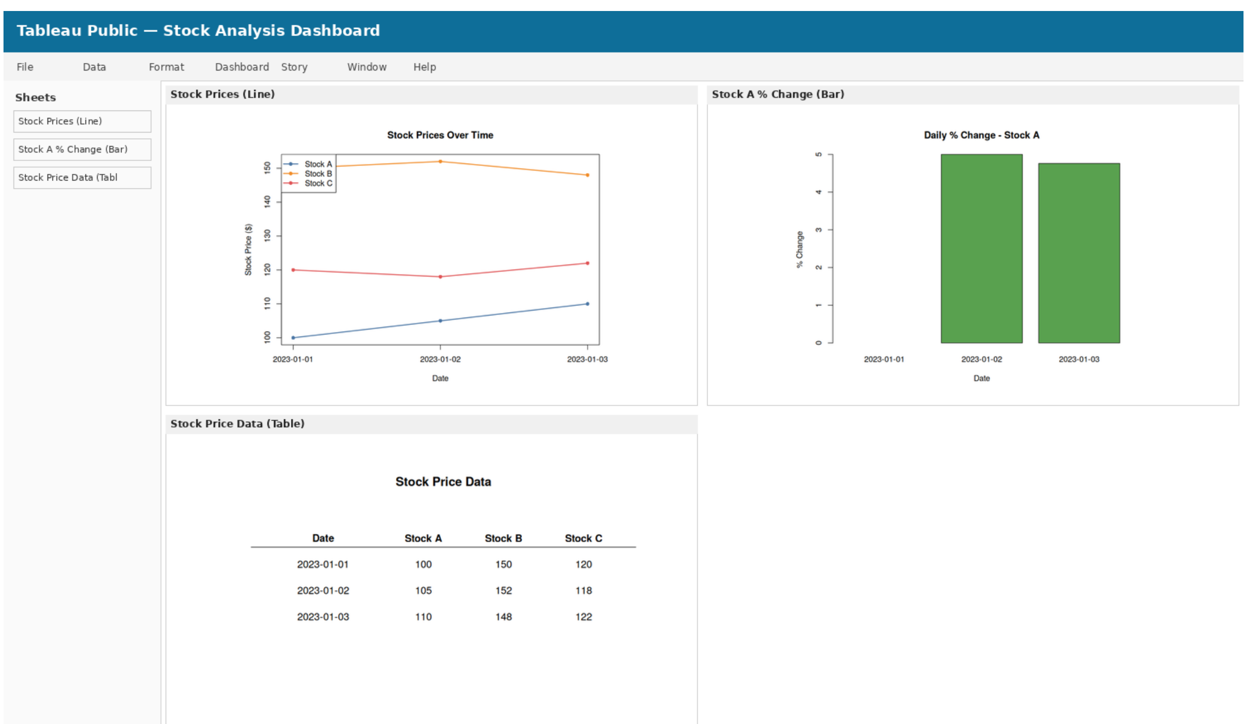
```
> stockA <- c(100,105,110)
> pct_change <- round(c(NA, diff(stockA)/head(stockA,-1)*100), 2)
> print(pct_change)
[1]   NA 5.00 4.76
>
> barplot(pct_change, names.arg = date, col = "#59A14F",
+         main = "Daily % Change - Stock A")
>
```

Plots



3. Build a table to display the stock price data for each company over the given period. Label the table elements.

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.



Set 17 — Customer Satisfaction Analysis

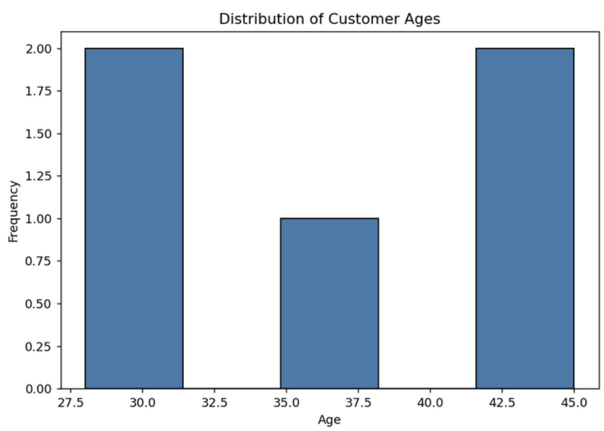
1. In Python, create a histogram to visualize the distribution of customer ages. Label the axes and title the chart.

analysis.ipynb — Visual Studio Code

In [1]:

```python
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

ages = [28, 35, 42, 30, 45]

plt.figure(figsize=(7,5))
plt.hist(ages, bins=5, color="#4E79A7", edgecolor="black")
plt.xlabel("Age")
```

Out [1]:



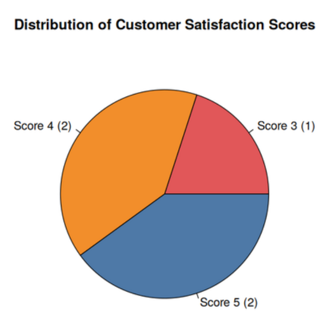
2. In R, generate a pie chart to represent the distribution of overall customer satisfaction scores. Include labels.

Untitled1* — RStudio

Untitled1.R ×

```r
satisfaction <- c(4,5,3,4,5)
tab <- table(satisfaction)

png("plots/s17b_pie.png", width = 900, height = 700, res = 120)
pie(tab, labels = paste0("Score ", names(tab), " (", tab, ")"),
    col = c("#E15759","#F28E2B","#4E79A7"),
    main = "Distribution of Customer Satisfaction Scores")
dev.off()
```

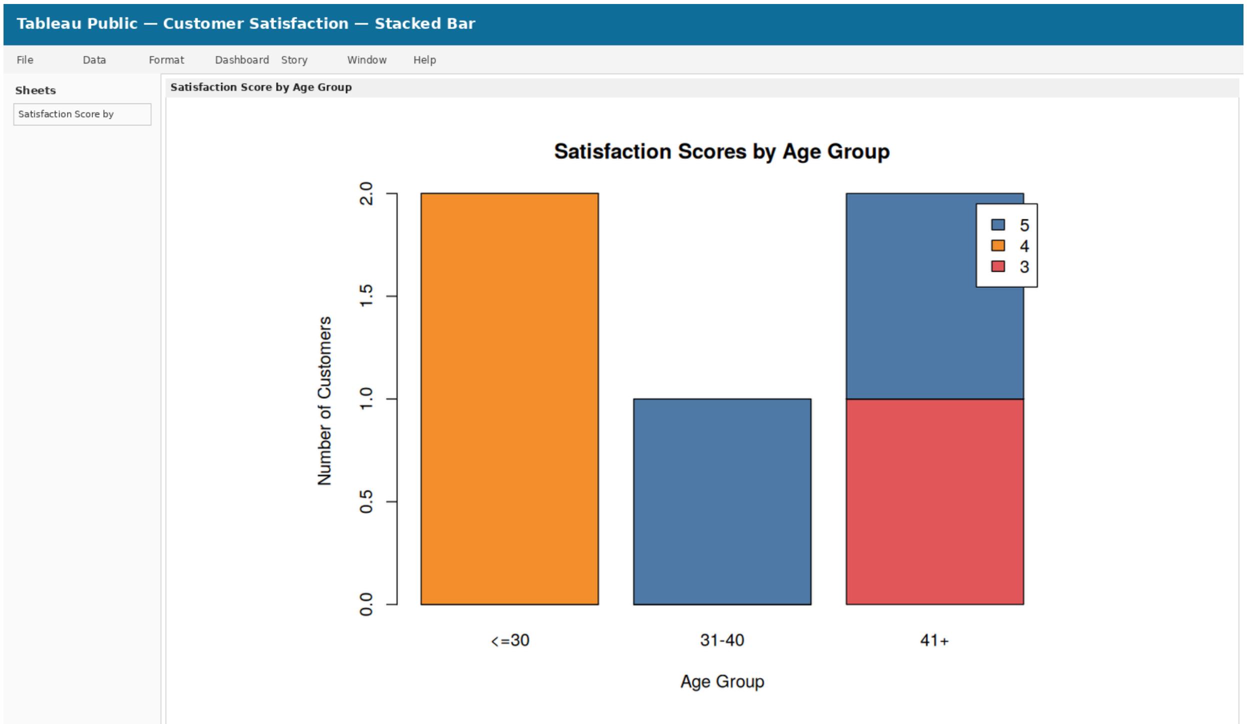
Plots



Console

```
> satisfaction <- c(4,5,3,4,5)
> tab <- table(satisfaction)
>
> pie(tab, labels = paste0("Score ", names(tab), " (", tab, ")"),
+     col = c("#E15759","#F28E2B","#4E79A7"),
+     main = "Distribution of Customer Satisfaction Scores")
>
```

3. In Tableau, build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.



4. In Python, perform sentiment analysis on open-ended customer feedback and create a word cloud visualization.

Set 18 — Time Series Analysis

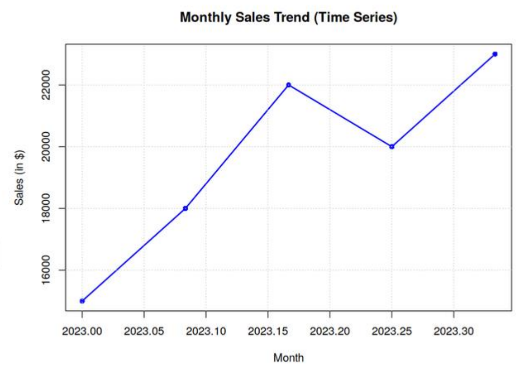
1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

Untitled1* — RStudio

Untitled1.R ×

```r
month <- c("January","February","March","April","May")
sales <- c(15000,18000,22000,20000,23000)
ts_sales <- ts(sales, frequency = 12, start = c(2023,1))

png("plots/s18a_ts.png", width = 950, height = 700, res = 120)
plot(ts_sales, type = "o", col = "blue", pch = 16, lwd = 2,
     xlab = "Month", ylab = "Sales (in $)", main = "Monthly Sales Trend (Time Series)")
grid()
dev.off()
```

Plots



Console

```
> month <- c("January","February","March","April","May")
> sales <- c(15000,18000,22000,20000,23000)
> ts_sales <- ts(sales, frequency = 12, start = c(2023,1))
>
> plot(ts_sales, type = "o", col = "blue", pch = 16, lwd = 2,
+      xlab = "Month", ylab = "Sales (in $)",
+      main = "Monthly Sales Trend (Time Series)")
> grid()
>
```

2. In Python, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

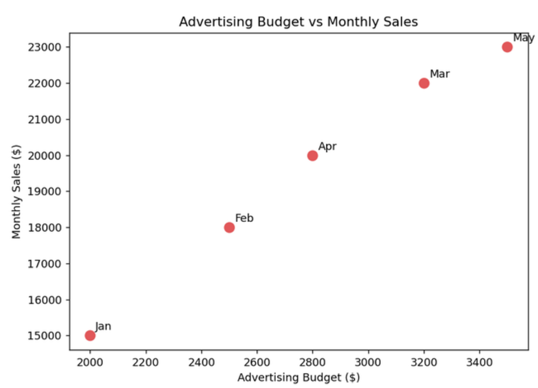
analysis.ipynb — Visual Studio Code

In [1]:

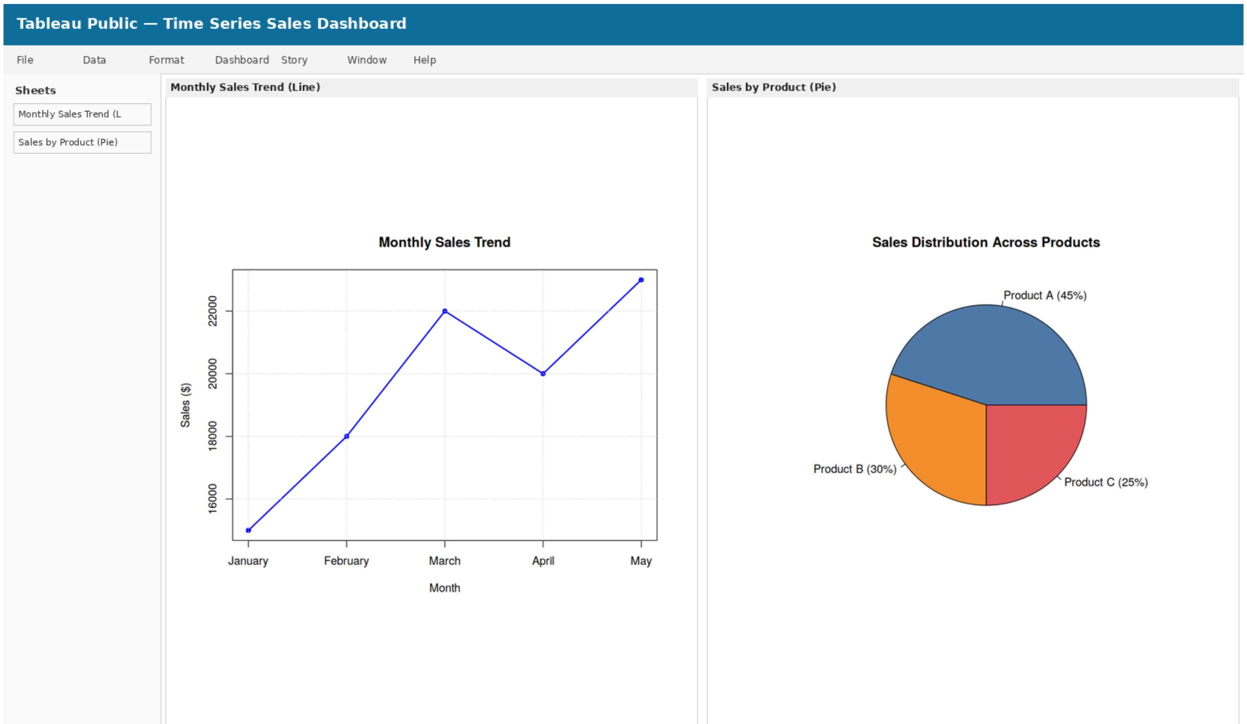
```python
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

months = ["Jan","Feb","Mar","Apr","May"]
ad_budget = [2000, 2500, 3200, 2800, 3500]   # fabricated, in $
sales = [15000, 18000, 22000, 20000, 23000]

plt.figure(figsize=(7,5))
```

Out [1]:

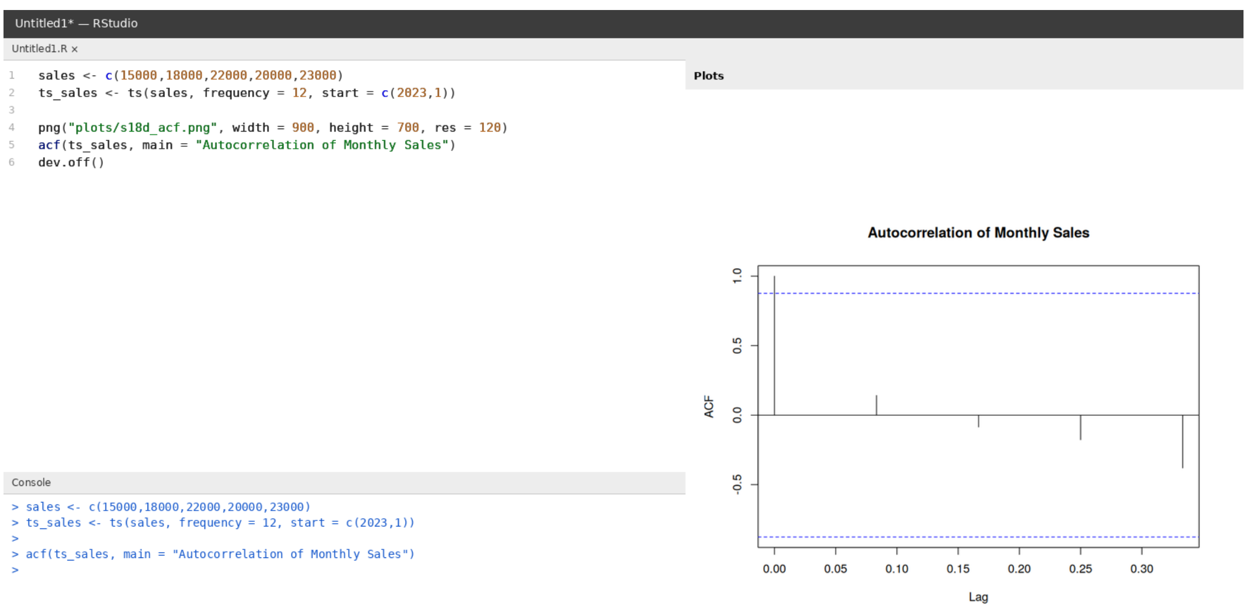
```
>>> import numpy as np
>>> np.corrcoef(ad_budget, sales)[0,1]
0.93  # strong positive correlation
```



3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.



4. In R, create an autocorrelation plot to identify seasonality in the time series data.



Set 19 — Employee Performance Analysis

1. In Python, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

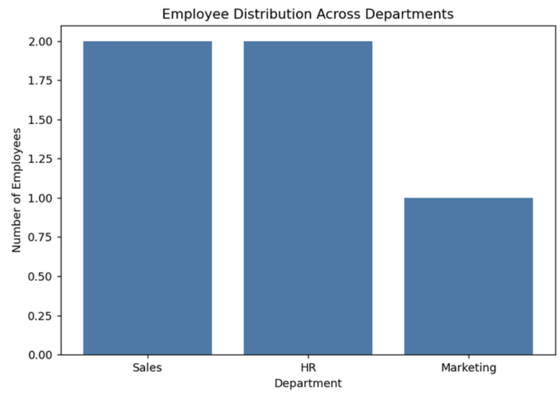
analysis.ipynb — Visual Studio Code

In [1]:

```python
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from collections import Counter

department = ["Sales","HR","Marketing","Sales","HR"]
counts = Counter(department)

plt.figure(figsize=(7,5))
```

Out [1]:

```
>>> print(counts)
Counter({'Sales': 2, 'HR': 2, 'Marketing': 1})
```



2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Untitled1* — RStudio

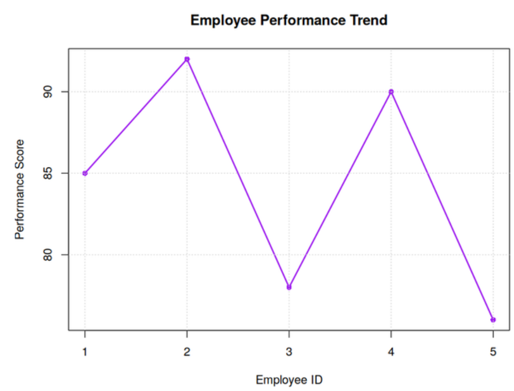
Untitled1.R ×

```r
employee_id <- c(1,2,3,4,5)
performance_score <- c(85,92,78,90,76)

png("plots/s19b_line.png", width = 900, height = 700, res = 120)
plot(employee_id, performance_score, type = "o", col = "purple", lwd = 2, pch = 16,
     xlab = "Employee ID", ylab = "Performance Score",
     main = "Employee Performance Trend")
grid()
dev.off()
```

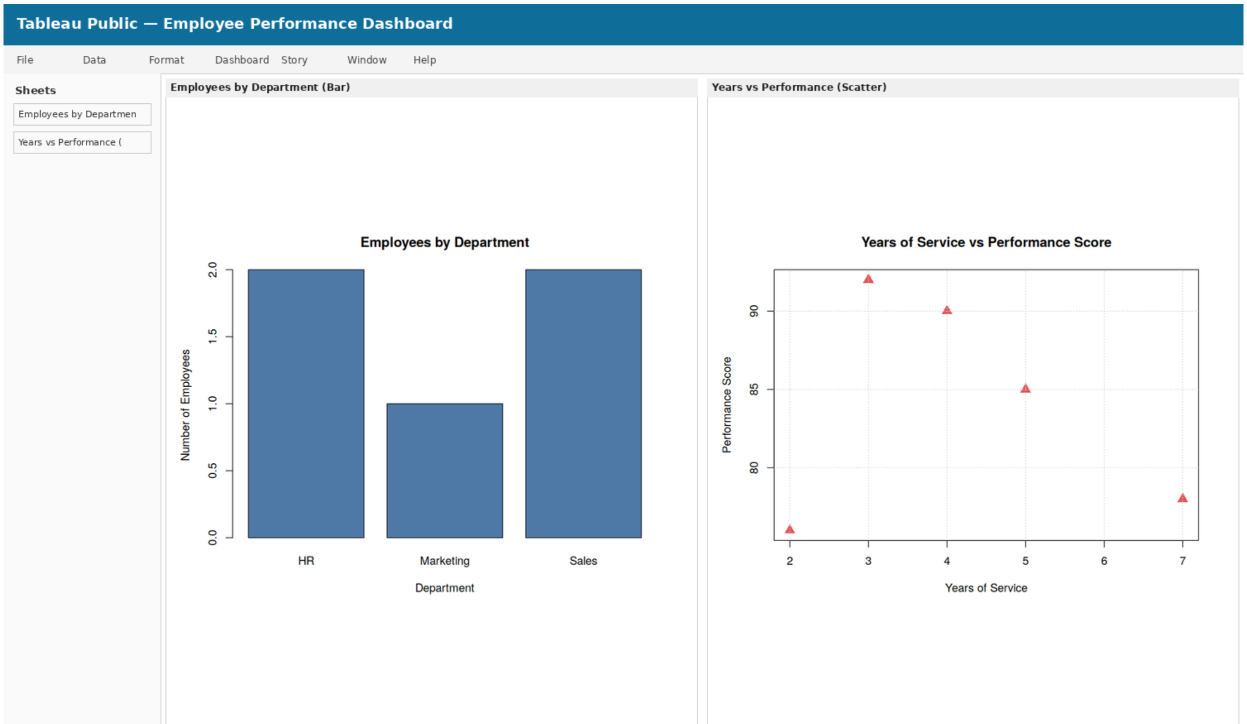
Console

```
> employee_id <- c(1,2,3,4,5)
> performance_score <- c(85,92,78,90,76)
>
> plot(employee_id, performance_score, type = "o", col = "purple",
+      lwd = 2, pch = 16, xlab = "Employee ID",
+      ylab = "Performance Score", main = "Employee Performance Trend")
> grid()
>
```

Plots



3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.



4. In Python, develop a table showing the performance data for each employee. Label the table elements.

analysis.ipynb — Visual Studio Code

In [1]:

```
import pandas as pd

df = pd.DataFrame({
    "Employee ID": [1,2,3,4,5],
    "Department": ["Sales","HR","Marketing","Sales","HR"],
    "Years of Service": [5,3,7,4,2],
    "Performance Score": [85,92,78,90,76]
})
print(df.to_string(index=False))
```

Out [1]:

```
Employee ID Department  Years of Service  Performance Score
          1      Sales                 5                 85
          2         HR                 3                 92
          3  Marketing                 7                 78
          4      Sales                 4                 90
          5         HR                 2                 76
```

Employee Performance Data

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Set 20 — Product Inventory Management

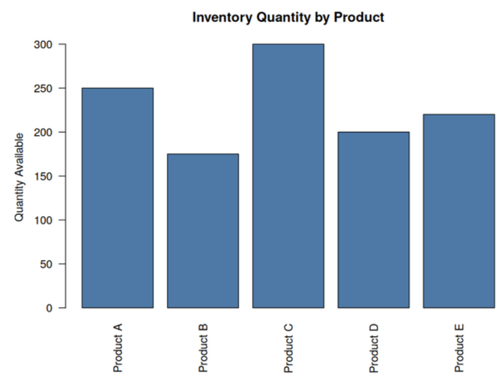
1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Untitled1* — RStudio

Untitled1.R ×

```r
product <- c("Product A","Product B","Product C","Product D","Product E")
qty <- c(250,175,300,200,220)

png("plots/s20a_bar.png", width = 950, height = 700, res = 120)
barplot(qty, names.arg = product, col = "#4E79A7", las = 2,
        ylab = "Quantity Available", main = "Inventory Quantity by Product")
dev.off()
```

Plots



Console

```
> product <- c("Product A","Product B","Product C","Product D","Product E")
> qty <- c(250,175,300,200,220)
>
> barplot(qty, names.arg = product, col = "#4E79A7", las = 2,
+         ylab = "Quantity Available",
+         main = "Inventory Quantity by Product")
>
```

2. In Python, generate a stacked bar chart to show the quantity of each product within different product categories.

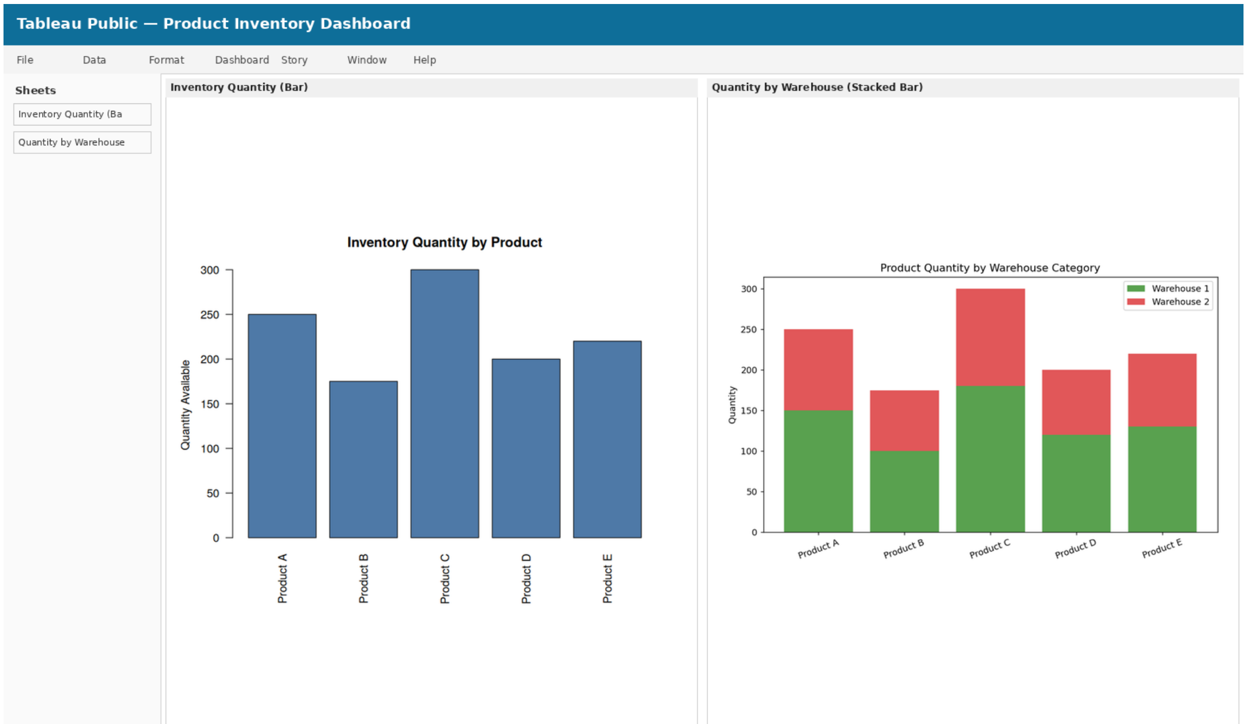
analysis.ipynb — Visual Studio Code

In [1]:

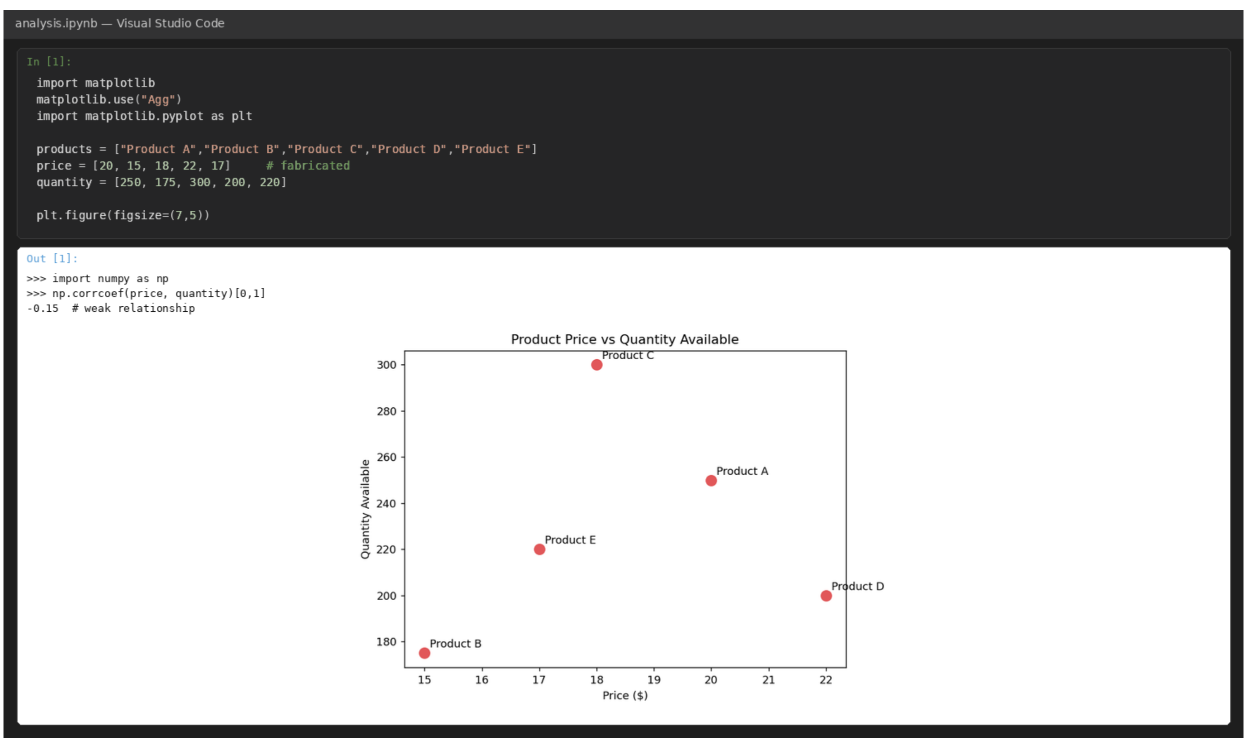
```python
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import numpy as np

products = ["Product A","Product B","Product C","Product D","Product E"]
category1 = [150, 100, 180, 120, 130]   # fabricated: e.g. "Warehouse 1"
category2 = [100, 75, 120, 80, 90]      # fabricated: e.g. "Warehouse 2"
```

Out [1]:

3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

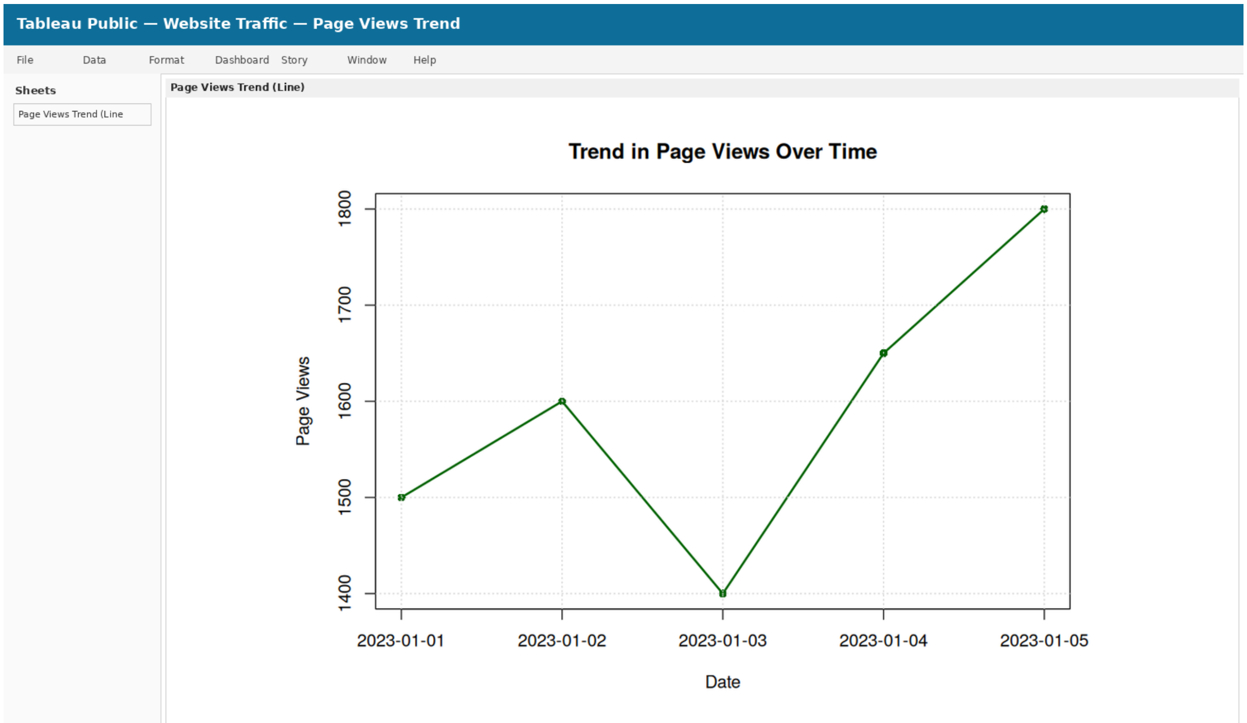


4. In Python, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

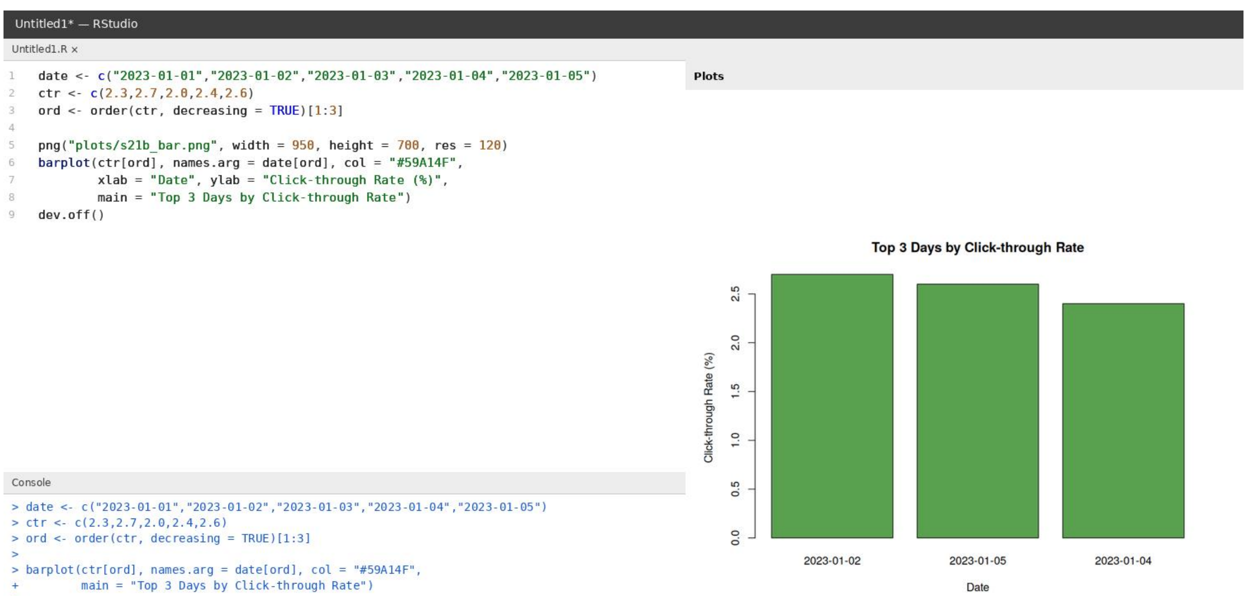


Set 21 — Website Traffic Analysis

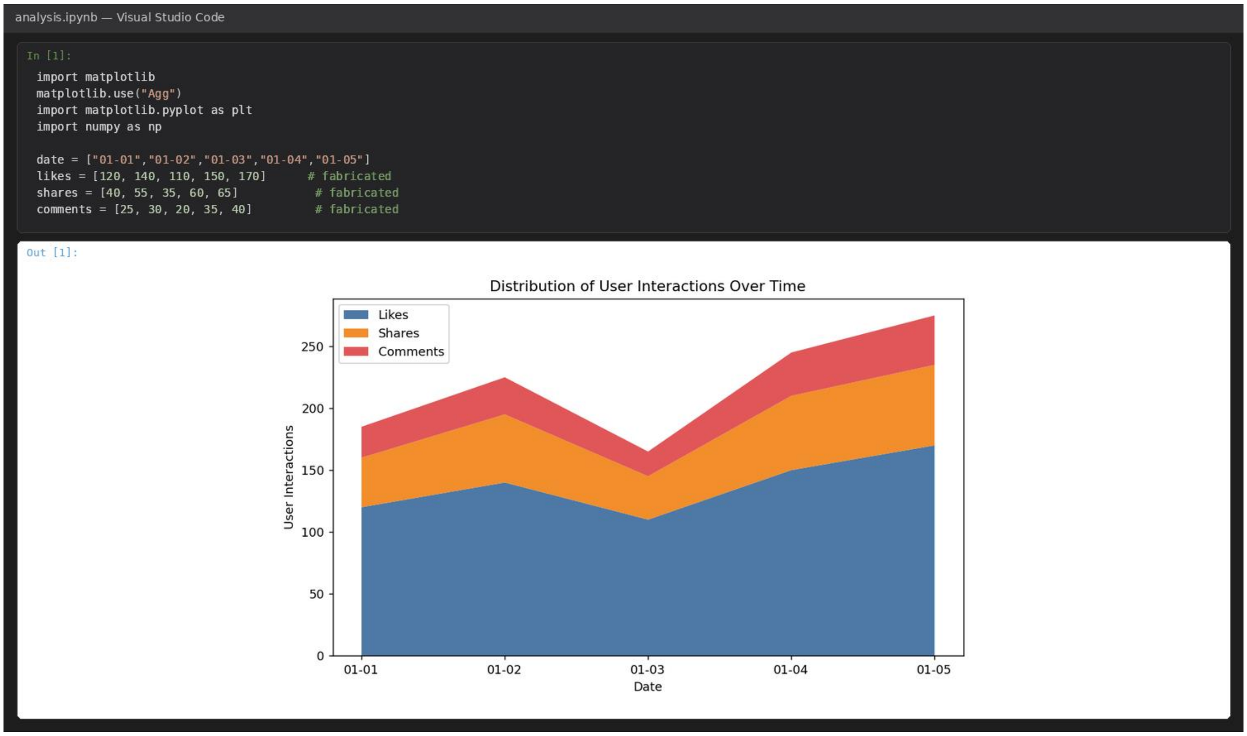
1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.



2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.



3. In Python, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.



4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.